

A High-Resolution TDC Design Based on Multistep Fine Time Measurement by Utilizing Delay-Adjustable Looped Carry Chains on FPGAs

Ke Cui¹, Jintao Yu¹, Jiaxuan Zou¹, and Xiangyu Li¹

Abstract—High-resolution and multichannel time-to-digital converters (TDCs) are needed in many electrical application systems. A new multistep fine time measurement method is proposed in this work, which is based on the delay-adjustable looped carry chains to obtain both high resolution and measurement precision. Each sub-step of the fine time measurement is assigned with a different measurement resolution in the descending order by dynamically configuring the delay line length. The time residue to be measured in the following sub-step becomes smaller, so that a finer resolution can be chosen. The overall oscillation number is the summation of the oscillation number of each sub-steps. The resolution of the TDC is determined by the last sub-step. So both high-resolution and small oscillation number (corresponding to high measurement precision) can be obtained. The TDC circuit was implemented on Stratix III field programmable gate array (FPGA) chip. Test results showed that the obtained resolution was less than 10 ps, and the measurement precision root mean square (rms) was about 13 ps. The differential nonlinearity (DNL) error and the integral nonlinearity (INL) error were in the range of -1 least significant bit (LSB) to 1 LSB without any bin calibration. The resource cost per TDC channel includes the lookup table (LUT) cost of 1064 and register cost of 318. The obtained TDC is comprehensively good considering its low resource cost and high linearity, and it is promising for multichannel TDC design.

Index Terms—Carry chain, field programmable gate array (FPGA), high linearity, multichannel TDC, ring oscillator (RO), time-to-digital converter (TDCs).

I. INTRODUCTION

LIGHT detection and ranging (LiDAR) applications in industrial automobile vehicles call for multichannel, low-power, and high-resolution time-to-digital converters (TDCs). The TDCs also act as core components in many other electrical application systems, such as fluorescence life-time imaging (FLIM), positron emission tomography (PET),

all-digital phase-locked loops (PLLs), high-energy physics, and quantum communication.

There are two platforms to implement the TDC circuits: one is the application specific integrated circuit (ASIC) and the other one is the field programmable gate array (FPGA) [1]. Due to the high design freedom of the delay lines, the ASIC-based TDC generally can obtain high performance, including ultrahigh resolution, low-power consumption, and small nonlinearity error. However, the development time of the ASIC-based TDC is long and the redesign cost is high. The off-the-shelf ASIC-based TDC may not comply with the actual system requirement of the TDC circuits. Such as for the LiDAR application, very dense channel number is required but a resolution of about 50 ps is enough. Commercially available TDC for such requirement may not exist, since most TDCs generally focus on high resolution and the channel number is small. On the other hand, FPGA-based TDC has very fast prototyping period and can be modified according to various application requirements. Modern FPGAs have huge logic resources and can build even hundreds TDCs in a single chip. However, due to the limited hardware resource type, the circuit architecture of the FPGA-based TDC is limited and generally leads to poorer performance compared with the ASIC-based TDC.

Typical TDC design methods include: the tapped delay line (TDL) [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17] [18], the pulse shrinking [19], [20], [21], and the Vernier delay line [22], [23], [24]. The measurement resolution is determined by the delay line resolution in the TDL, the propagation delay difference of the two delay lines in the Vernier method, and the delay difference of the rising and falling edges in the pulse shrinking. During the past years, although different FPGA resources are proposed to realize the delay line, such as look-up tables (LUTs) and digital signal processing (DSP) blocks, the carry chains show more fascinating potential to construct the TDCs. Compared with the LUT-based delay element, the carry chain has smaller delay value and more regular structure which can be synthesized by software without manual intervention. Compared with the DSP, the carry chain resource widely exists among the entire FPGA chip and can be used to construct multichannel TDC with different resolution requirements.

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Among various FPGA-based TDC circuits, the TDL-based method using carry chain as the delay line is the dominant technique. The circuit is easy to be implemented, and owing to the ultra-fine resolution of the carry chain of latest high-end FPGAs, the TDL-based TDC can achieve very fine resolution of smaller than 10 ps. Useful techniques to improve the measurement resolution contain wave union [10], [13], multichain structure [6], [12], and delay matrix [18]. However, due to the intrinsic large delay variations of the carry chain, the realized nonlinearity error is usually large, being about 3–5 least significant bits (LSBs). Many methods have been proposed to solve the problem such as bin-by-bin calibration [7], bin decimation [3], automatic bin calibration [14], and tuned-TDL method [2]. However, these methods complicate the TDC design and are not friendly for multichannel design since the corrections should be carried out independently for each channel.

In 2017, our group first introduced the carry chain to construct the Vernier delay line by connecting the end of a carry chain to its front to form the ring oscillator (RO) [22]. In the previous work, two carry-chain-based ROs with similar structure have proven to hold very close delay value. By adjusting the length of the RO, the specific delay of the Vernier delay line can be easily controlled. The recently developed delay-controllable scheme makes the TDC design process to be conducted using software without manual intervention to the compiled TDC circuit [23]. This greatly reduces the TDC design complexity.

However, the resolution of the TDC is limited to about 30 ps. The measurement precision is inversely proportional to the root square of the RO oscillation number [24], [25], so the resolution cannot be set too small to avoid large oscillation number during the measurement. Another drawback is the long dead time, which is determined by $T_s \times n$, where T_s represents the oscillation period, and n represents the total oscillation number. In the previous design, the dead time usually turns out to be 400–600 ns.

This article proposes a novel TDC architecture called the multistep fine time measurement method to improve the resolution and lower the dead time. The work is built on the dynamically delay-adjustable looped carry chain structure. The basic idea is to exploit the resolution-adjustable attribute of the carry-chain-based RO. The fine time measurement process contains several sub-steps, with each sub-step being assigned with different measurement resolutions in the descending order. The total oscillation number can be greatly decreased in this way, and the effective resolution of the TDC is determined by the resolution of the final sub-step. It is worthwhile to mention that the proposed TDC shares some similarity with the so-called successive approximation TDC [26], [27]. The successive approximation TDC successively compares the fine time interval with a variable reference time delay and determines the TDC result from the most significant bit (MSB) to the LSB. The proposed TDC uses the delay difference (Vernier effect) of two delay lines to interpolate the fine time and contains several sub-steps holding different delay resolutions. Both TDC implementation methods need an effective method to construct the delay line of variable delay values.

The TDC circuit was implemented on a Stratix-III FPGA chip. The FPGA chip is identical to those in the works [22], [23], [24] to make the performance comparison reasonable. Experimental results show that the obtained resolution was less than 10 ps, and the measurement precision root mean square (rms) was about 13 ps. The differential nonlinearity (DNL) error and the integral nonlinearity (INL) error were in the range from -1 to 1 LSB without any bin calibration. The logical consumption is that the LUT cost is 1064 per TDC channel and the register cost is 318 per TDC channel. This demonstrates that our design can outstand the TDL-based TDCs for multichannel TDC applications.

II. DESIGN METHOD

A. Circuit Operation Principle

To cover large measurement range and maintain high resolution, two measurement steps are adopted, including one coarse step and one fine step. The coarse step uses a coarse counter to record large time interval. The fine step uses the RO-based Vernier delay line which is constructed from the looped carry chains to record the fine time within one coarse clock period. The measurement result is represented by the coarse counter value and the fine counter value. The overall TDC structure is shown in Fig. 1(a). It consists of four modules: the clock extraction module, the coarse counter module, the fine time measurement module, and the time assembler module. The coarse counter module and fine time measurement module generate the corresponding coarse and fine timestamps. The clock extraction module extracts out the fine time interval between the hit signal and the nearest clock signal after it. The time assembler module combines the coarse and fine timestamps together and outputs the final TDC result.

Unlike the previous works [22], [23], [24], the fine step in this work is further divided into several sub-steps by assigning each sub-step with a different measurement resolution. This is feasible since the adopted Vernier delay line structure can be dynamically adjusted with different delay line lengths. The RO structure is shown in Fig. 1(b). It consists of a looped carry chain with total length of N . Each output of the delay element is connected to one input port of the N -channel multiplexer. The channel selection port of the multiplexer can be dynamically configured to decide the actually used delay elements along the carry chain. By assigning different channel selection parameters to the two delay lines, the measurement resolution (equaling to the delay difference of the two delay lines) of the Vernier delay line can be dynamically controlled. It should be noted that the propagation delay of the carry chain delay line part consists of two components: the delay along the carry chain and the delay from the end of the carry chain element to the multiplexer output. The first delay increases as the number of the carry chain element increases, but the second delay is unpredictable after each circuit compilation (for the Intel Quartus software). This would lead to the phenomenon that the overall propagation delay may decrease as the number of the selected carry chain element increases [23]. However, the design goal of the TDC is to find two looped carry chain with small propagation delay difference. This will not

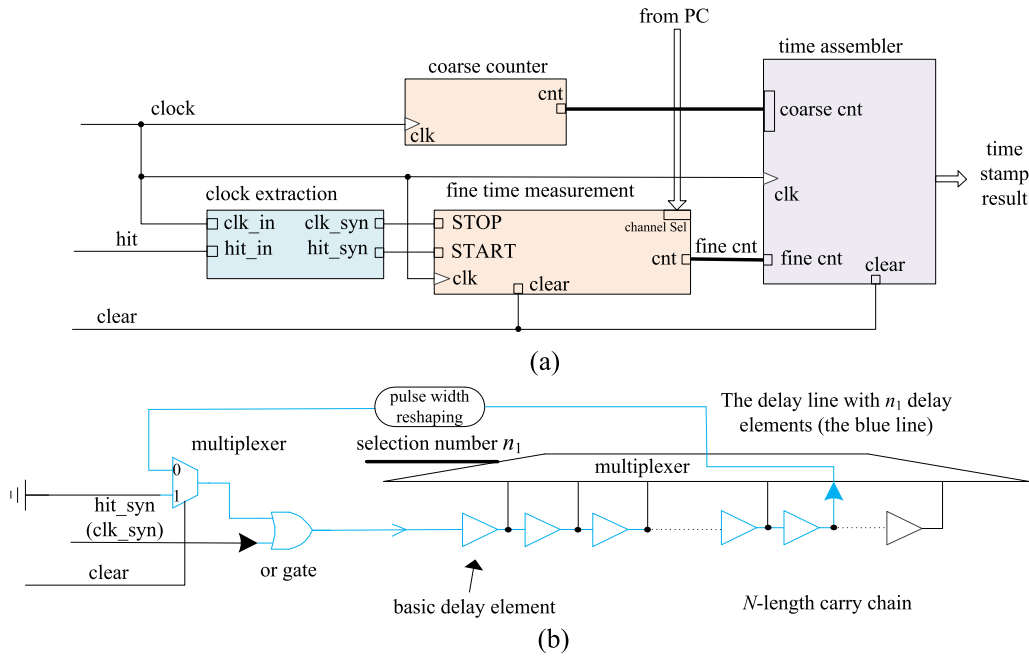


Fig. 1. TDC structure. (a) Overall composition. (b) Looped carry chain with dynamically delay-adjustable capability.

be affected by the above-mentioned phenomenon. According to our experimental discovery in the previous works, the delay-adjustable RO can provide a wide resolution range from smaller than 10 ps to several hundreds of picoseconds. The mentioned delay line can be synthesized automatically by the software and its delay resolution can be controlled without manual intervention to the complied circuit result. It provides great design freedom for the TDC circuits with least design efforts.

The working process of the fine time measurement step is depicted in Fig. 2(a). Initially, the leading event is represented by the START signal and the lagging event is represented by the STOP signal. The time delay between them is initially Δt_1 , and satisfies $0 < \Delta t_1 < T_{\text{pos}}$, where T_{pos} represents the positive pulsewidth of the START signal. The two signals propagate along the two looped ROs of the Vernier delay line. In the first sub-step, the channel selection parameter set ($\text{Sel}_1, \text{Sel}_2$) is configured, where Sel_1 controls the multiplexer of the START signal and Sel_2 controls the multiplexer of the STOP signal. The ($\text{Sel}_1, \text{Sel}_2$) makes the Vernier delay line to have the resolution of r_1 , which should be smaller than the coarse clock period but is relatively larger than the final target resolution of the TDC. For example, the r_1 can be chosen to be several hundred picoseconds to guarantee that the STOP signal can exceed the START signal only after several times of oscillation. Then the relative phase between the START and the STOP changes to the case 2 in Fig. 2(a). Now the STOP signal is ahead of the START signal with a finer time amount of Δt_2 , which is in the range of $0 < \Delta t_2 < r_1$. During case 2, the channel selection parameter set is changed to the new value of ($\text{Sel}_3, \text{Sel}_4$), which makes the Vernier delay line to have the new resolution of r_2 . Here r_2 is negative to let the START signal to catch up the STOP signal after several times of oscillation. The absolute value of r_2 can be

chosen to be several times smaller than r_1 (several tens of picoseconds for example), so that case 2 also only undergoes several times of oscillation. At the beginning of case 3, the START signal becomes ahead of the STOP signal, just owing the same phase relationship as case 1 but with a much smaller delay amount. Then the channel selection parameter set is changed to ($\text{Sel}_5, \text{Sel}_6$) with the choice of the resolution r_3 to be a positive value within the range of $0 < r_3 < -r_2$. The Vernier delay line works in the same way as the process in case 1 until the STOP signal exceeds the START signal again. Similarly, the sub-steps of the fine time measurement can be carried out furthermore with the resolution choice of $0 < r_{2i+1} < -r_{2i}$, and $0 < -r_{2i} < r_{2i-1}$, where r_i represents the resolution in the i th sub-step. When the sub-index is odd, the resolution is positive. When the sub-index is even, the resolution is negative. Let f_j represents the fine counter of the oscillation number in the j th sub-step, then the fine time can be calculated in the following way:

$$t_f = \sum_{j=1}^k r_j \times f_j \quad (1)$$

where k represents the total number of the sub-step in the fine time measurement.

The corresponding timing diagram of the proposed TDC is shown in Fig. 2(b). The coarse counter runs freely. When the coarse clock detects the high-level status of the hit signal, the corresponding counter value is recorded as the coarse value. Then the fine time interval between the hit signal and the nearest successive clock signal is extracted out, which is further measured by the Vernier delay line. During the fine time measurement, the hit signal corresponds to the START signal and the clock signal corresponds to the STOP signal. The fine time measurement contains several sub-steps.

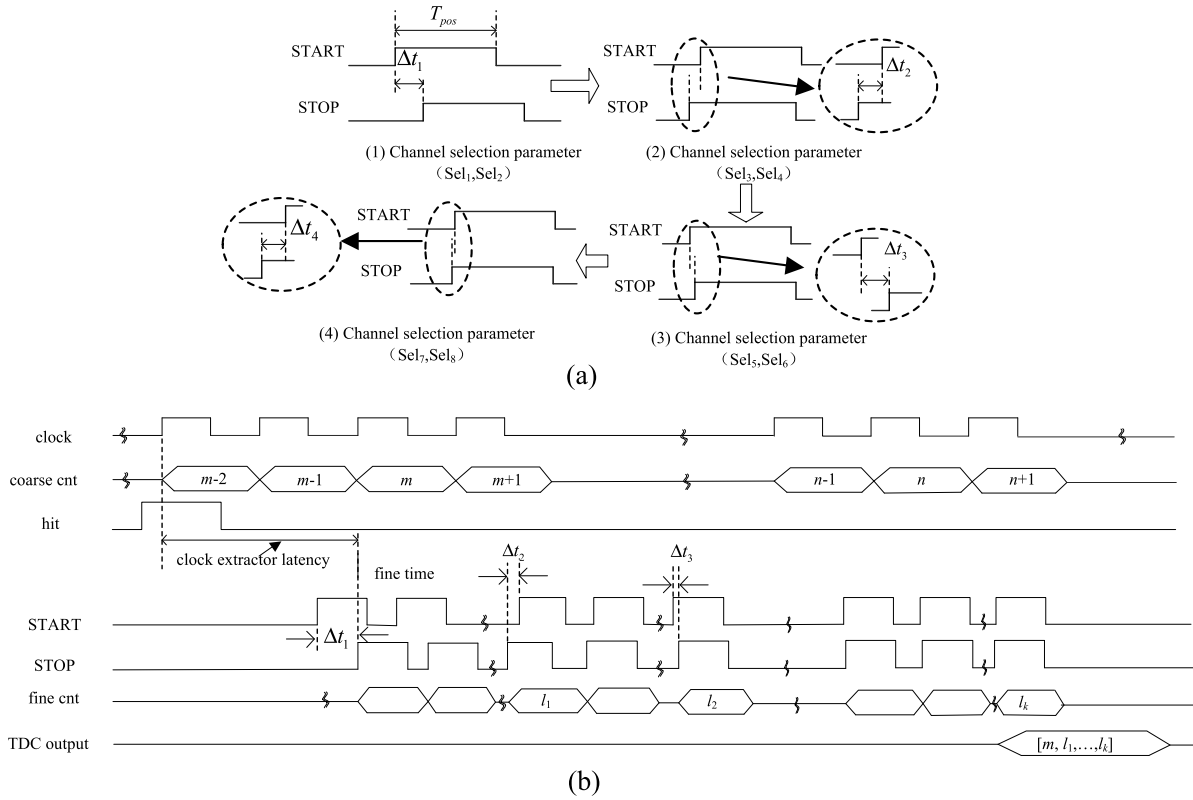


Fig. 2. Working principle of the TDC. (a) Working process of the fine time measurement. (b) Timing diagram.

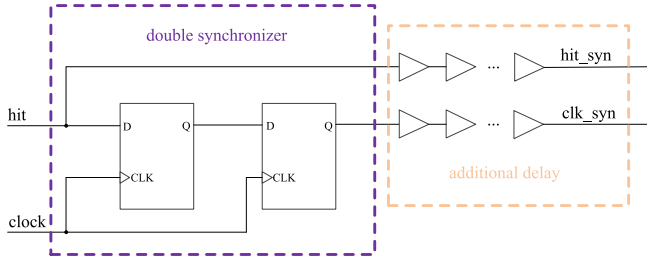


Fig. 3. Circuit of the clock extraction module.

As shown in Fig. 2(b), in the end of the k th sub-step, the fine counter value l_k is recorded. The switching of the sub-steps is determined by the relative phase transition of the START and the STOP signal as illustrated above. It should be noted that since the fine counter is not cleared in the end of each sub-step, the fine counter gives the accumulative result of the sub-steps. The fine counter value is calculated as, $f_1 = l_1$ and $f_i = l_i - l_{i-1}$ ($i = 2, 3, \dots, k$). In the end of the k th sub-step, the overall time measurement is finished, and the combined timestamps ($m, l_1, l_2, \dots, l_k$) are generated as the measurement result, where m represents the recorded coarse counter value.

B. Circuit Implementation

In this section, the specific circuit design of Fig. 1(a) is illustrated. The circuit of the fine time extraction module is shown in Fig. 3. It contains the double synchronizer part and the additional delay part. The double synchronizer uses two

D-type flip-flops (DFFs) to deal with metastability problem. The coarse clock signal is connected to the clock port of the DFFs, and the hit signal is connected to the D port of the first DFF. After detecting the hit signal and the small propagation delay of the DFF, the Q port turns to high, which is synchronous to the coarse clock signal. The second DFF is inserted to solve the metastability problem when the active edge of the clock signal is very close to the edge of the hit signal. Then the hit signal and the extracted clock signal can be fed into the following Vernier delay line. However, the route delay of the hit signal to the START signal may be different from that of the extracted clock signal to the STOP signal. If the route delay of the hit signal is much longer, this incurs the wrong initial leading time relationship of the STOP signal over the START signal, which violates the required timing requirement in Fig. 2. A delay chain of cascaded delay buffers is inserted between the hit (clock) signal and the START (STOP) signal. In our circuit design, even number of LUT-based inverters are applied. The inverter amounts are adjusted to guarantee that the STOP signal is always behind the START signal when reaching the Vernier delay line.

The Vernier delay line circuit is shown in Fig. 4. It contains two looped carry chains. The upper delay line is designed to propagate the START signal and the below delay line is designed to propagate the STOP signal. According to the working principle of the TDC, the STOP signal needs to sample the START signal continuously. This is realized by adopting a DFF as the phase detector. The STOP signal is connected to the clock port of the DFF while the START signal is connected to the D port of the DFF. After each oscillation,

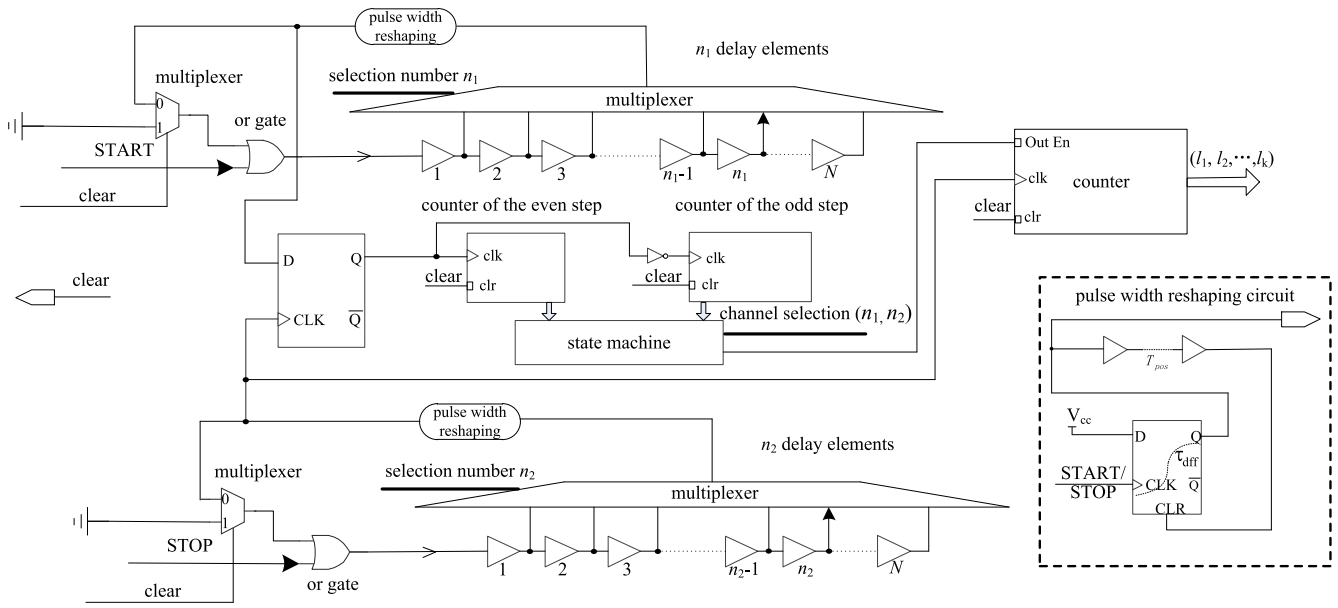


Fig. 4. Structure of the switchable Vernier delay line.

the STOP samples the status of the START signal. The Q port of the DFF is labeled as the STATUS signal. For the odd (even) number of the sub-steps, the high-to-low (low-to-high) transition can be detected in the end of the sub-step. The state machine module in Fig. 4 synthesizes the status of the sub-steps and controls the switching of the channel selection parameters. It uses two counters to discriminate the number of the sub-steps. One counter is triggered by the rising edge of the STATUS signal and denotes the $2n$ th sub-step. The other counter is triggered by the falling edge of the STATUS signal and denotes the $(2n - 1)$ th sub-step. The fine counter is triggered by the STOP signal to run upward continuously. The fine counter values in the end of each sub-step are recorded. After the k th sub-step is finished, the fine counter and the Vernier delay line are cleared to make itself ready for the next fine time measurement.

C. Metastability Problem of the Fine Time Measurement

The fine time extraction module (Fig. 3) adopts two consecutive DFFs to overcome the sampling metastability problem. However, the phase detection circuit in Fig. 4 only uses one DFF. This can incur the sampling metastability problem if the STOP signal is very close to the START signal. Since the additional DFF introduces a clock latency, Fig. 4 cannot apply the two-DFFs method. If two DFFs are used for the phase detector, the remaining fine time residue becomes $\Delta t_{i+1} + T_{osc}$ in the $(i + 1)$ th sub-step (here T_{osc} represents the oscillation period of the STOP signal), which will largely increase the oscillation time in the i th sub-step. This diminishes our attempt to adopt the multistep fine time measurement method to decrease the overall oscillation number.

Instead, we propose to use two TDC channels with different initial delays between the START and STOP signals to overcome this problem. The additional delay is inserted between the hit signal and the data input port of the clock

extraction module. This additional delay changes the initial phase relationship between the START and STOP signal. If the delay value is small for one channel, the delay value should be large for the other channel. At least one TDC channel would not face the sampling metastability problem in this way. Clearly, the optimal value of the delay is equal to half of the coarse clock period T_c . The delay value can be reflected by the fine counter value f_2 in the second sub-step. By choosing the corresponding value f_1 which owns larger f_2 , the sampling metastability problem for the first sub-step is overcome. Similarly, if the sampling problem for the i th sub-step needs to be resolved, the f_i corresponding to the channel result having larger f_{i+1} should be chosen. The sampling metastability problem can be perfectly eliminated if the selected design parameters of the TDC satisfies the following conditions:

$$(\Delta d \bmod T_c) \approx 2/T_c \quad (2)$$

$$(\Delta d \bmod r_i) \approx r_i/2 \quad (3)$$

where Δd represents the inserted additional delay between the two TDC channels.

D. Design Process

In Section II-A, a general way to implement the multistep fine time measurement TDC scheme is described. Although increasing the sub-step number can improve the TDC performance, accurately generating the required resolution of each sub-step is difficult for FPGA implementation. In the specific TDC circuit realized in this work, three sub-steps are designed, but the obtained result is very satisfying as demonstrated in Section III. In our design, the target resolution of r_1 is set to several hundred picoseconds, the target resolution of r_2 is set to several tens of picoseconds, and the target resolution of r_3 is set to around 10 ps.

To implement the proposed TDC, the following design steps are adopted as suggested in our previous works [22], [23], [24]. It contains the TDC construction process and the channel selection parameter determination process.

In the TDC construction process, to ease the channel selection parameter determination, the two differential delay lines should have identical structure. This is accomplished by the intellectual property (IP) importation method. The delay line structure is firstly synthesized and fit in an independent project. It is then exported out as an IP file. Then in the TDC design project, the two delay lines are generated by importing the IP file which keeps the original routing information. The Quartus software can automatically and intelligently remain the relative placement and routing circuit inside the delay line, but the absolute physical position of each delay line is made different. So, the software can automatically add some constant position displacement to any logic unit of the delay line so that all TDC channels still hold the uniform circuit structure. In the TDC construction process, it is also important to adjust the amount of the delay inverters as shown in Fig. 3. This delay only needs to be adjusted once before using the TDC circuit. This can be realized by using the manual editing tool such as the engineering change order (ECO) tool provided by Quartus software. The advantage of the ECO tool is that it only modifies the locally re-edited circuit without changing the remaining circuits so the delay of the Vernier delay line keeps constant during this process.

In the channel selection parameter determination process, different channel selection parameters are configured into the fine time measurement module. The ends of the delay lines are introduced to an external oscilloscope, and the oscillation period corresponding to each channel selection parameter can be recorded. Finally, three channel selection parameter collections which own several hundred picoseconds, several tens of picoseconds, and about ten picoseconds can be sorted out to realize our proposed TDC circuit.

III. RESULT

A. Test Setup

The TDC circuit was designed by using the Verilog language, and automatically synthesized by the Quartus software. The overall carry chain length for each delay line is 32. It was implemented on the Stratix III FPGA chip as the same with our previous works [22], [23], [24] to prove the performance gain from the proposed circuit architecture. The FPGA was hosted on a self-made printed circuit board (PCB) which contains the universal serial bus (USB) 2.0 communication interface to transfer the TDC results to the PC. The TDC structure and the experimental setup is shown in Fig. 5. Since our TDC records the timestamps, two TDC channels are designed to obtain the time interval test results. The time interval is equal to the timestamp difference between the two TDC channels. Each TDC channel contains two Vernier delay lines to address the metastability problem as illustrated in Section II. Since each Vernier delay line contains two carry-chain-based ROs, there are totally eight ROs in our design. And the used carry chain length is 128 per TDC channel. The

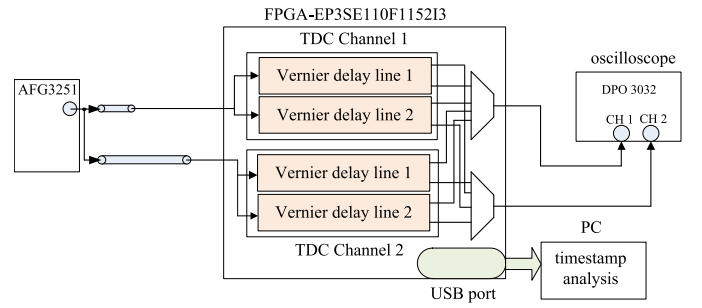


Fig. 5. Experimental setup.

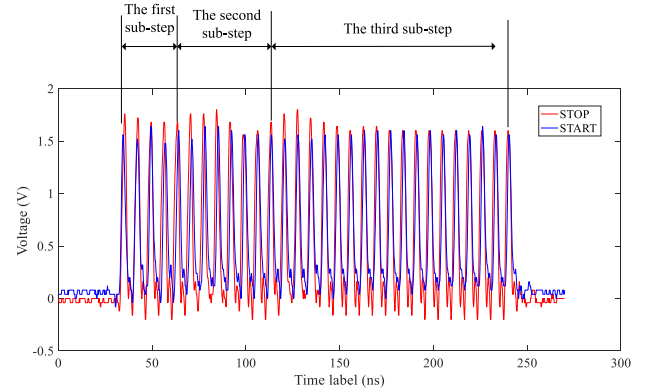


Fig. 6. Actual operation waveform of the Vernier delay line.

clock rate of the coarse counter is 600 MHz (corresponding to the clock period of 1667 ps). The clock is generated from the ON-chip PLL of the FPGA. The reference clock of the PLL is from an OFF-chip 20 MHz quartz crystal. The width of the coarse counter is set to 10 bits to make sure that it can work stably under the 600 MHz frequency. The corresponding operating range of the TDC is then $1.667 \text{ ns} \times 2^{10} = 1707 \text{ ns}$. The coarse counter and the fine counter value sequences are combined and transferred to PC for further analysis. The hit signal is generated from an arbitrary function generator (AFG3251 by Tektronix Company) whose clock is uncorrelated with the TDC clock. The hit signal is split into two paths with different cable lengths (corresponding to different time intervals) before reaching the FPGA board. An external oscilloscope (DPO 3032 by Tektronix Company) was used to observe the oscillation period of each delay line in the TDC. The oscillation period under different channel selection parameters were tested and recorded, so that proper channel selection parameter set can be determined for each sub-step measurement. The nominal voltage supply of the FPGA chip was 1100 mV and the tests were carried out under the ambient temperature of 20 °C.

B. Working Waveform

After testing the actual oscillation period of each delay line of the designed TDC, the proper channel selection parameter sets were determined to form the multistep fine measurement. Fig. 6 gives one of the actual working waveforms which is sampled by the oscilloscope after the TDC circuit was correctly configured. The blue line represents the START

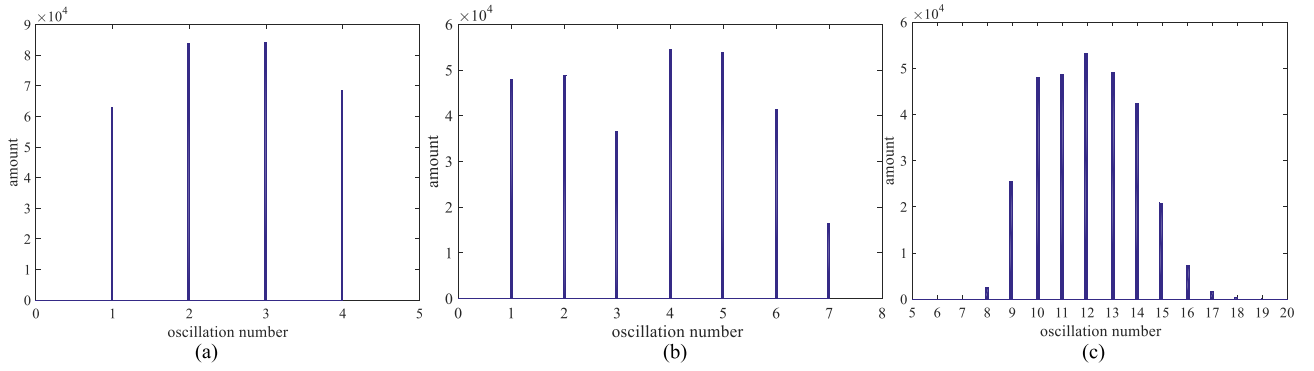


Fig. 7. Oscillation numbers in each of the fine time measurement sub-steps. (a) First sub-step. (b) Second sub-step. (c) Third sub-step.

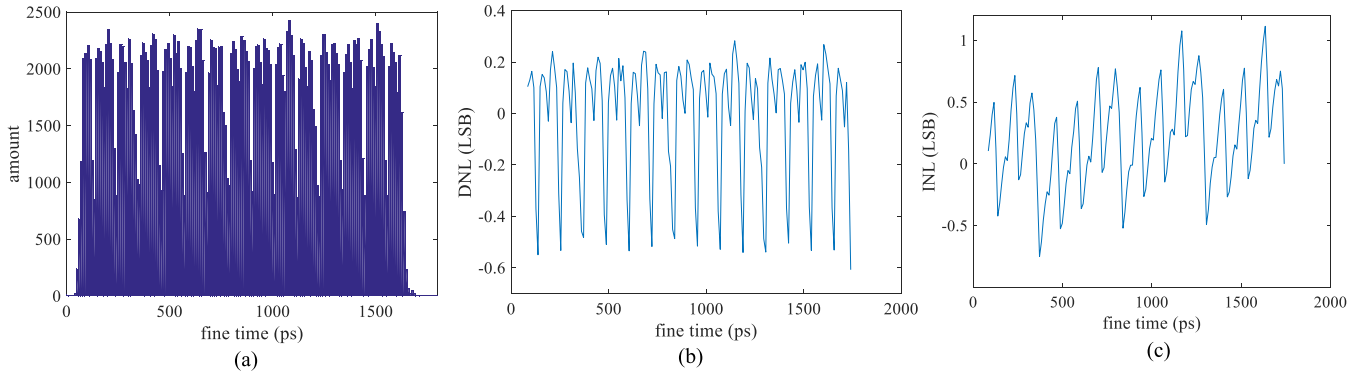


Fig. 8. DNL and INL of one of the delay line of the proposed TDC. (a) Fine time bin distribution. (b) DNL curve. (c) INL curve.

signal and the red line represents the STOP signal. It obviously contains three sub-steps. In the first sub-step, the red line oscillates faster than the blue line. After four times of oscillation, the STOP signal transcends the START signal and the channel selection parameter set is changed to the new value. In the second sub-step, the START signal oscillates faster than the STOP signal. After seven times of oscillation, the START signal transcends the STOP signal and the channel selection parameter set is changed to the third value. In the final step, the STOP signal oscillates faster than the START signal once more but with a much finer resolution. When the STOP signal transcends the START signal for the second time, the overall fine time measurement step ends, and the three fine time counter values in each end of the sub-steps are recorded and outputted. In Fig. 6, the recorded fine counters are $f_1 = 4$, $f_2 = 7$, $f_3 = 16$, so the fine time interval is calculated as $4 \times r_1 + 7 \times r_2 + 17 \times r_3$ according to (1). Here the resolutions in each sub-step are determined by analyzing the oscillation number distributions as illustrated in Section III-C.

C. Performance Evaluation

To reduce statistical error, more than 200 000 hits are accumulated to evaluate the bin (or the fine counter value) distribution. The oscillation number distributions of each sub-step of one example delay line in the TDC circuit are depicted in Fig. 7. Fig. 7(a) shows the bin distribution of the first sub-step, in which the oscillation number spans from 1 to 4. The minimal oscillation number is caused by the additional delay

from the clock extraction module to the fine time measurement module. The average resolution of the first sub-step is calculated as $1667/4 = 416.7$ ps. Fig. 7(b) shows the bin distribution of the second sub-step, in which the oscillation number spans from 1 to 7. The corresponding resolution is calculated as $416.7/7 = 59.6$ ps. As for the third sub-step, the oscillation number corresponding to the full width at half maximum (FWHM) is used, and the oscillation range is selected from 9 (left half value) to 15 (right half value). The side value and the corresponding width of the sidelobe are caused by the limited measurement precision. Due to this reason, we do not use the oscillation number that have very small oscillation number (from 8 to 18). Using the different choices of the oscillation number span, the calculated TDC resolution and the measurement precision rms may vary slightly. It is validated that our choice can make the measurement precision rms smallest. So the resolution of the third step is calculated as $59.6/(15 - 9 + 1) = 59.6/7 = 8.5$ ps. The TDC resolution is determined by the third sub-step and equals to 8.5 ps.

The timestamps of the obtained hit signals from the above Vernier delay line are calculated to obtain the fine time distribution by using (1). The fine time distribution with the bin width of 8.5 ps is depicted in Fig. 8(a). The corresponding DNL is calculated and depicted in Fig. 8(b) and the INL is depicted in Fig. 8(c). The DNL is in the range from -0.6 to 0.3 LSB and the INL is in the range from -0.8 to 1 LSB. The DNL and INL of all other Vernier delay lines in our design also lie in the range from -1 to 1 LSB. Unlike the TDL-based TDC, the DNL and INL of our TDC method do

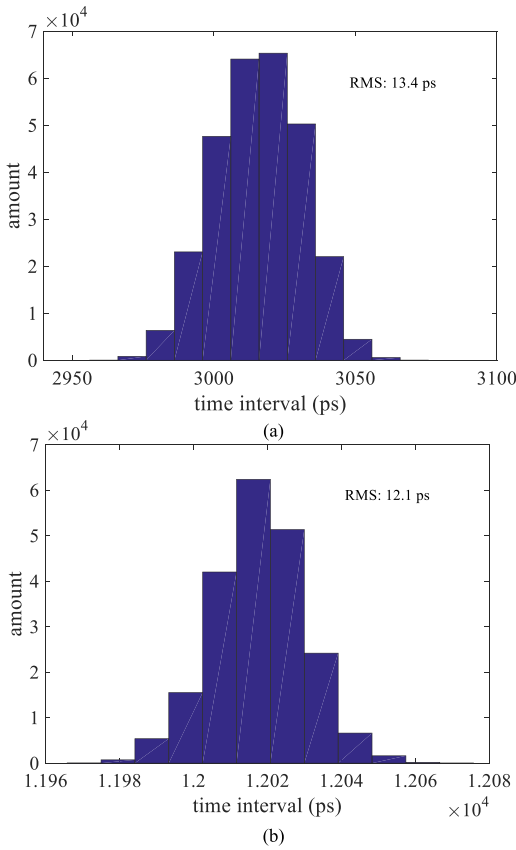


Fig. 9. Histograms of the time interval measurement. (a) Mean time of 3025 ps corresponding to the short delay cable. (b) Mean time of 12010 ps corresponding to the long delay cable.

not need any complicate calibration algorithm, which greatly saves the resource cost. This makes the proposed TDC very beneficial for multichannel implementation.

The resolutions of other delay lines turn out to lie in the range of about 7–10 ps, and the INLs and DNLs all lie in the range from -1 to 1 LSB. It should be noted that the linearity is dependent on the specific hit event since the measurement result is combined from two Vernier delay lines. Statistically, the nonlinearity error can be represented by the average of the nonlinearity error of the delay lines in the TDC circuit. Since all the tested delay lines have the DNL and INL in the range from -1 to 1 LSB, it is certain that the nonlinearity error of the TDC circuit is still in the range from -1 to 1 LSB.

Two TDC channels were used to test the time interval measurement performance. One short cable and one long cable were used to generate two different time intervals. The start hit signal was measured by TDC channel 1 and the stop hit signal was measured by TDC channel 2. The time interval was obtained by subtracting the timestamp of TDC channel 1 from that of TDC channel 2. The histograms of the time intervals are depicted in Fig. 9. Fig. 9(a) gives the histogram of the short cable which shows the mean time interval of 3025 ps with the measurement precision rms of 13.2 ps. Fig. 9(b) gives the histogram of the long cable which shows the mean time interval of 12010 ps with the measurement precision rms of 12.1 ps.

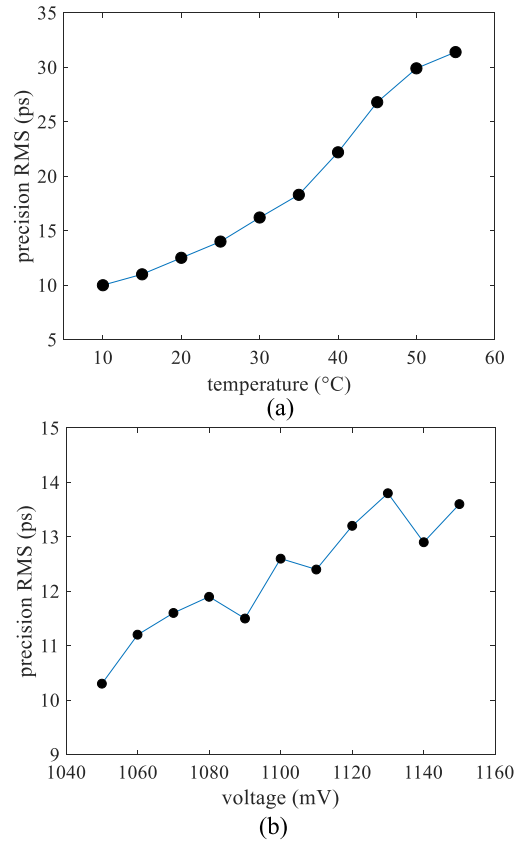


Fig. 10. TDC performance versus the PVT variations. (a) Measurement precision rms versus the temperature variation. (b) Measurement precision rms versus the voltage variation.

The TDC circuit is susceptible to process, voltage, and temperature (PVT) variation, so the measurement precision versus the ambient temperature and the voltage variation was studied. To do the temperature stability test, the PCB board was placed in a temperature control chamber and the ambient temperature was controlled to change from 10°C to 55°C with the step of 5°C . The core voltage was maintained to 1100 mV during the test. Fig. 10(a) shows the result of the measurement precision rms versus the temperature variation. The precision rms degrades as the temperature increases, probably due to the increased circuit noise under high ambient temperature. To do the voltage stability test, the core voltage supplied to the FPGA chip was controlled to change from 1050 to 1150 mV with the step of 10 mV. The ambient temperature was maintained to 20°C . Fig. 10(b) shows the result of the measurement precision rms versus the core voltage variation. The precision rms becomes slightly larger under higher voltage, but the degeneration effect is not severe. So maintaining lower operating temperature is more important to guarantee the measurement precision of the TDC circuit.

The resource cost per TDC channel from the Quartus report is as following. LUT cost of 1064 and register cost of 318. Three main reasons contribute to the low resource cost of the proposed TDC. The first reason is that the used carry chain length is short being 128 per TDC channel. The second reason is that the TDC directly outputs the timestamp result without any decoding module (unlike the thermometer code

TABLE I
PERFORMANCE COMPARISONS WITH OTHER RECENTLY DEVELOPED FPGA-BASED TDCS

Works	Year	Chip	Method	Resolution (ps)	Precision RMS (ps)	DNL (LSB)	INL (LSB)	LUT usage	Register usage
[3]	2016	UltraScale	TDL bin decimation	2.97	4.20	-1.00~3.95	-1.50~1.95	664	1652
[6]	2017	Virtex-7	TDL Multichain	1.15	3.5	-0.98~3.50	-5.90~3.10	15255	N/S
[7]	2016	Virtex-6	TDL Dual Phase	10	12.83	-1.00~1.88	-1.63~3.93	N/S	N/S
[13]	2022	UltraScale	Sub-TDL Wave Union	2.48	3.63	-0.93~1.68	-1.78~2.67	2460	3463
[14]	2022	Zynq-7000	Sub-TDL Wave Union Auto Calibration	9.83	13.86	-0.14~0.16	-0.25~0.42	764	1095
[17]	2021	Artix-7	TDL Sliding Scale Moving Average	4.88	<8.03	-0.10~0.15	-0.23~0.28	2962	4157
[18]	2020	Stratix-IV	TDL PLL Delay Matrix	15.60	15.60	-0.18~0.18	-0.016~0.14	9886	N/S
[19]	2018	Actel-A2F200M3F	Two-Stage Loop Shrinking	8.5	42.4	-0.22~0.36	-0.61~0.91	N/S	N/S
[22]	2017	Stratix-III	RO-based Vernier line	31.0	36.1	-0.08~0.07	-0.09~0.09	319	104
[24]	2020	Stratix-III	Bidirectional RO-based Vernier line	24.5	28.0	-0.20~0.25	0.03~0.82	986	172
This work	-	Stratix-III	Multi-step RO-based Vernier line	<10	13.2	-1.00~1.00	-1.00~1.00	1064	318

N/S=not specified

to fine timestamp conversion module in the TDL-based TDC). The third reason is that the obtained bin distribution is rather uniform, and no complex calibration algorithm is needed. This demonstrates that our design can obtain both high resolution and high linearity with very low resource cost. The maximal oscillation period in each sub-step is 8 ns, and the maximal total oscillation number is $4 + 7 + 17 = 28$ according to Fig. 6. The largest dead time is then $8 \text{ ns} \times 28 = 224 \text{ ns}$.

Performance comparisons between this work and other recently developed FPGA-based TDCs are listed in Table I.

IV. CONCLUSION

To reduce the oscillation number of the recently proposed TDC based on the carry-chain-based RO Vernier delay line, this work proposes the multistep fine measurement method. By lending the benefit of the delay-adjustable attribute of the carry-chain-based RO, several fine measurement sub-steps can be realized with descending resolution. The overall oscillation number is equal to the summation of the oscillation number in each sub-step, which is significantly reduced compared with the traditional uniform resolution method. Both the resolution and measurement precision rms can be realized at the 10 ps level. Compared with the dominant TDL-based TDC, our method has relatively small nonlinearity error, so no bin width calibration algorithm is needed under most cases. This greatly reduces the TDC design complexity and saves the resource cost. The proposed method is very promising to realize multi-channel and high-resolution TDC on a single FPGA chip.

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